

Combating Mutuality with Difficulty Factors in Multi-class Imbalanced Data: A Similarity-based Hybrid Sampling

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Abstract—Multi-class imbalanced problem widely exists in real-life applications and has been a challenging issue. Existing sampling methods including decomposition approaches and dedicated approaches have limitations in handling the complex mutual relationships along with data difficulty factors. Actually, the relative minorities are critical in mutual relationship, and the data difficulty factors are harmful for these minority classes. In this paper, we propose *SHSampler*, a similarity-based hybrid sampling to combat the mutuality by addressing data difficulty factors in multi-class imbalanced data. Specifically, *SHSampler* firstly utilizes a sample similarity and dissimilarity estimation to identify data difficulty factors. Then, *SHSampler* conducts a relative majority weakening undersampling and a relative minority strengthening oversampling to reduce the negative impact of data difficulty factors and highlight the importance of the minorities. Extensive experiments over 20 typical datasets demonstrate the superiority of *SHSampler* in terms of MAUC and mGM when compared with 6 state-of-the-art methods.

Index Terms—multi-class, imbalanced data, sampling, classification, sample similarity, data difficulty factors

I. INTRODUCTION

Imbalanced data learning becomes one of the most crucial tasks in machine learning and data mining [1]. The scenario of imbalanced data appears when sample sizes of different classes are uneven, where the dominated classes are called majority and the rare classes are called minority. Usually, the minority are more nontrivial and informative than the majority [2]. Conventional classification algorithms, guided by the standard accuracy, show deficiency in imbalanced data, manifested in bias towards the majority and ignorance of the minority [3].

Existing researches have made substantive progress in imbalanced learning [4], but most of them concern the binary imbalanced data. Actually, multiple classes are often present in many real-life applications such as text categorization [5] and human activity recognition [6]. Compared with binary scenarios having a specific majority and minority, multi-class imbalanced data suffer from complex mutual relationships due to different sizes of multiple classes [7]. Also, it encounters data difficulty factors including small disjuncts, noises and overlaps with multiple classes [8]. These characteristics make multi-class imbalanced classification more challenging.

The state-of-the-art methods can be mainly categorized into two groups: (1) *Decomposition methods* [9], [10], [11], [12]. They generally transform a multi-class problem into several binary-class sub-problems, such as OVO (one-vs-one) [11] and OVA (one-vs-all) [12]. These methods have the

possibility of adopting full-fledged sampling techniques and classifiers [13], but may lead to some data regions and the mutual relationships among classes being left unlearned [14]. (2) *Dedicated methods* [15], [16], [17], [18], [19], [20]. These methods intuitively sample the original multi-class imbalanced data to overcome the dilemma of skew class distribution. Some of them sample the specific class individually to make its number reach a fixed criteria, without considering other classes [15], [16]. Others aim to obtain the relationship of one class to all other classes [17] or to the relative majorities [18], [19], generating and cleaning samples in specific regions. Besides, a few dedicated methods consider the neighboring class interrelations for sample removing and duplication [20]. Dedicated methods have superiority in general applications due to independence on classifiers. However, they mostly ignore the varying degrees of effects from different classes, making the information among classes deficiently studied [8].

The above two groups have limitations in efficiently handling the mutual relationships to emphasis the important classes. Moreover, data difficulty factors have more negative effect on the minorities than the majorities, ultimately deteriorating the recognition of minorities [8]. Consequently, it is necessary to find a research avenue to combat the mutual relationships by addressing the data difficulty factors. To this end, we propose a similarity-based hybrid sampling (*SHSampler*) for multi-class imbalanced data. To handle the mutual relationships and mitigate the effect of data difficulty factors, we answer two questions:

(i) **How to identify data difficulty factors including overlap, noise and small disjuncts in diversified data distribution?** The data difficulty factors intervene in or interweave with other classes, which makes them hard to be identified. To this end, we implement a sample similarity and dissimilarity estimation to distinguish safe samples and data difficulty factors. That is, we identify the samples having high dissimilarity to its class and low similarity to other classes, which means these samples are far away from the same class and close to the different classes, respectively.

(ii) **How to utilize the characteristic of data difficulty factors in mutual relationship to emphasis the important minority classes?** Data difficulty factors play varying roles on different classes because of imbalanced class sizes. To highlight the importance of the relative minorities, we design a relative majority weakening undersampling and a relative

minority strengthening oversampling. That is, we target to eliminate data difficulty factors of the relative majorities and supplement safe samples for the relative minorities.

To summarize, the contributions of this paper are as follows:

(1) We propose a sample similarity and dissimilarity estimation to distinguish data difficulty factors in multi-class imbalanced data.

(2) We design a relative majority weakening undersampling and a relative minority strengthening oversampling to highlight the relative minorities.

(3) Experiments on 20 typical benchmark datasets demonstrate that SHSampler achieves more accurate performance compared with 6 state-of-art methods.

The remainder of this paper is organized as follows. Section II describes the motivation. Section III gives the details of SHSampler. Section IV conducts experimental study to prove its effectiveness. Section V briefly reviews the related work. Section VI and VII remark our discussions and conclusions.

II. MOTIVATION

Besides the global imbalance ratio [8], multi-class imbalanced problem is intractable due to other several influence factors. In this section, we summarize these factors into two major categories: complex mutual relationships and data difficulty factors.

A. Complex Mutual Relationships

Different from binary-class imbalanced scenarios having specific majority and minority class, the relationships among multiple classes are more unobvious. A class may be majority when compared to some classes, but also may be minority or well-balanced for the others [21]. Besides, there are two cases called multi-minority and multi-majority, respectively. The former happens when one majority and multiple minorities exist while the latter happens when one minority and multiple majorities exist [7]. These two cases make the mutual relationships more obscure, which negatively affects the classification performance.

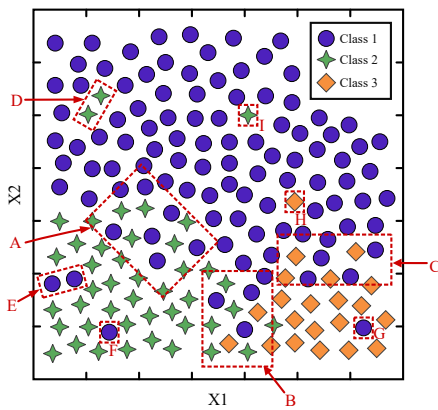


Fig. 1. Sketch map of imbalanced data with three classes.

We show a two-dimensional example with three classes in Fig. 1. For class 2, class 1 is majority while class 3 is minority;

for class 1, both class 2 and 3 are minority, and class 3 has a smaller sample size; for class 3, both class 1 and 2 are majority, and class 1 has a larger sample size.

B. Data Difficulty Factors

Data difficulty factors include class-overlap, small disjuncts and noise [22].

- **Class-overlap.** Class-overlap happens when different class samples distribute in a common area, and it may appear with more than two classes in multi-class scenarios [21]. Fig. 1 shows that regions A, B and C are overlapping area. Especially, part B consists of three different class samples, which brings more difficulties to classifiers.
- **Small disjunct and noise.** Small disjunct, as a conjunctive subconcept, is much error prone in classification [23]. Noise intervening in other class significantly impacts the classification performance [24]. Moreover, as the number of classes and the sizes of each class are various, it is difficult to identify the isolated sample as whether small disjunct or noise. For example, in Fig. 1, regions D and E are possible small disjunct of class 2 and 1, respectively. Samples F and G can be deemed as noises intervening in class 2 and 3. As class 2 and 3 have rare samples, samples I and H may be either small disjunct or potential noise.

In summary, complex mutual relationships and data difficulty factors pose difficulties in multi-class imbalanced learning. Actually, among the mutual relationships of different classes, the relative minorities deserve more attention than the majorities in real-life applications [2], [20]. For instance, in medical diagnosis, suspected patients can make further consultation but missed diagnosis may lead patients not promptly treated [25]. Furthermore, the recognition of the critical minority deteriorates more quickly than the intermediate or majority when disturbed by data factors [8]. Following these motivations, we propose to highlight the relative minority in mutual relationships by addressing the data difficulty factors.

III. SIMILARITY-BASED HYBRID SAMPLING

To address the challenges associated with multi-class imbalanced problem, we propose a novel similarity-based hybrid sampling (SHSampler) to combat the class mutuality by handling the data difficulties in multi-class imbalanced data. SHSampler is mainly based on the principle of highlighting the importance of the relative minorities. It firstly detects the data difficulty factors with a sample similarity and dissimilarity estimation process (§III.A). Then it resamples different classes to weaken the relative majority and strengthen the relative minority (§III.B and C). In this section, we describe the details of SHSampler.

A. Sample Similarity and Dissimilarity Estimation

As discussed above, data difficulty factors refer to the samples interweaving with different class samples or intervening in different class areas. Contrarily, the samples locating in the dense area of its class, and the samples far from different

classes are much safer. To distinguish the data difficulty factors, we propose to estimate the sample similarity and dissimilarity to classes.

Let class $A = \{x_1, x_2, \dots, x_i, \dots, x_{|A|}\}$, where $|\cdot|$ is the cardinality of a class. The compactness degree d , denoting how widely or narrowly samples of A are distributed, can be measured by the intra-class distance as follows:

$$d = \frac{\sum_{x_m \in A} d\left(x_m, \left(\sum_{x_n \in A} x_n\right) / |A|\right)}{|A|}, \quad (1)$$

where $d(*, *)$ denotes the Euclidean distance between two vectors. Thus given a sample $x_i \in A$, its dissimilarity to class A is defined as:

$$DI_A(x_i) = \sum_{x_j \in Set_{(x_i)}^{DI}} e^{d(x_i, x_j) - d} \cdot (d(x_i, x_j) - d), \quad (2)$$

where $Set_{(x_i)}^{DI} = \{x_j | j \neq i \wedge d(x_i, x_j) > d\}$. It can be seen that, samples locating in the dense area of its class have low DI , and samples locating far from the dense area of its class have high DI . We give Fig. 2 to illustrate the dissimilarity estimation process, in which the red stars represent the samples constituting $Set_{(x_i)}^{DI}$. The dissimilarity $DI(x_i)$ is smaller than $DI(x_j)$.

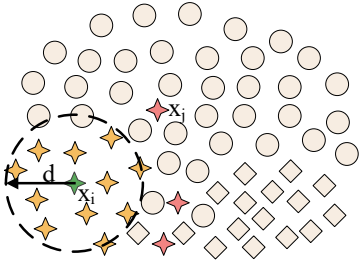


Fig. 2. Sample dissimilarity estimation process.

Identically, let class $B = \{y_1, y_2, \dots, y_i, \dots, y_{|B|}\}$. The separability degree m , denoting how far or closely samples of class A and B are distributed, can be measured by the inter-class distance as follows:

$$m = d\left(\left(\sum_{x \in A} x\right) / |A|, \left(\sum_{y \in B} y\right) / |B|\right). \quad (3)$$

Thus for the given x_i , its similarity to class B is defines as:

$$SI_{(A,B)}(x_i) = \sum_{y_i \in Set_{(x_i)}^{SI}} e^{\alpha - d(x_i, y_i)} \cdot (\alpha - d(x_i, y_i)), \quad (4)$$

where $Set_{(x_i)}^{SI} = \{y_i | y_i \in B \wedge d(x_i, y_i) < \alpha\}$, and $\alpha = d + m$. It can be seen that, samples intervening in other class have high SI , samples locating far from that class have low SI . We give Fig. 3 to illustrate the similarity estimation process. In Fig. 3(a), for estimating SI of x_i to the class denoted by circle, the red circles represent the samples constituting $Set_{(x_i)}^{SI}$. The similarity $SI(x_i)$ to that class is smaller than

$SI(x_j)$. In Fig. 3(b), for estimating SI of x_i to the class denoted by rhombus, the red rhombus represents the samples constituting $Set_{(x_i)}^{SI}$. The similarity $SI(x_i)$ to that class is smaller than $SI(x_k)$.

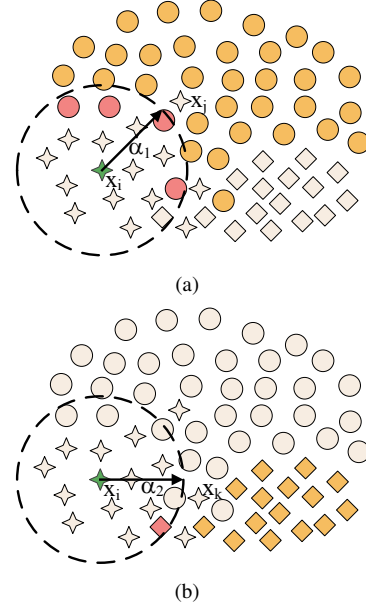


Fig. 3. Sample similarity estimation process.

Based on above, each sample has a DI to its class and has different SI s to other classes, respectively. We conclude that: (1) Safe samples have lowest DI and SI s. (2) Overlapping samples have high SI s to the interweaving classes. (3) Small disjuncts and noisy samples have high DI , and also have highest SI s to classes they intervene in. It is noted that when the samples of a class are distributed more dispersively, the isolated samples have lower DI than those belonging to a more compact class. (4) Redundant samples [26], which are far from the decision boundary and the dense area within the class, have high DI . Therefore, we can detect those data difficulty factors by estimating sample similarity and dissimilarity, and further tackle them to reduce their negative impact on multi-class imbalanced classification.

B. Relative Majority Weakening Undersampling

To reduce the global imbalance ratio, it is necessary to determine a fixed criteria to rebalance the dataset. Due to the probable multi-majority and multi-minority cases, only under-sampling the multiple majorities suffers from information loss, while only oversampling the multiple minorities may cause overfitting [7]. To make a compromise, we propose a hybrid sampling method by undersampling the classes whose sizes are above the average size of all classes and oversampling the classes below the average.

Furthermore, data difficulty factors of prevalent classes have negative impact on the rare classes, deteriorating the performance of standard classifiers. Also, as the various class sizes exists, these samples play varying roles towards different classes. The authors in [8] claim that data difficulty factors

such as overlap are more harmful for the class with smaller size. When overlap occurs, the smaller size a class has, the more quickly its recognition deteriorates. To this end, it is essential to weaken the negative impact of the relative majority, making the relative minority more visible.

Suppose a generic imbalanced dataset with N classes: $D = \{C_1, C_2, \dots, C_m, \dots, C_N\}$. Given classes C_m and C_n , if $|C_m| > |C_n|$, C_m is a relative majority to C_n , and C_n is a relative minority to C_m . For sample $x_i \in C_m$, its dissimilarity to C_m and similarity to C_n are denoted as $DI_{C_m}(x_i)$ and $SI_{(C_m, C_n)}(x_i)$, respectively.

Suppose $|C_m|$ is bigger than the average size, for sample $x_i \in C_m$, we define its undersampling weight as:

$$W(x_i) = \sum_{C_n \in D, m \neq n} \left(\frac{|C_m|}{|C_n|} \cdot SI_{(C_m, C_n)}(x_i) + DI_{C_m}(x_i) \right) \quad (5)$$

To reduce the size of C_m to the average, we employ a roulette wheel selection operator [27] to select the samples one by one through probabilities obtained by the undersampling weight. According to this operator, (1) the samples interweaving with the relative minority have priority to be eliminated than those interweaving with the relative majority; (2) the samples interweaving with the relative minority with a smaller size have priority to be eliminated than those interweaving with in the relative minority with a larger size; (3) redundant samples that have limited contribution to support the classifier and further increase the classification costs [26], also have possibility to be eliminated in our proposed strategy.

C. Relative Minority Strengthening Oversampling

Oversampling has shown apparent superiority in many publishes [28], which provides a higher proportion of safe samples and a lower proportion of unsafe samples. In multi-class imbalanced scenarios, oversampling techniques encounter more problems to determine the synthesis seed and the way of sample generation under the complex class relationships. To strengthen the importance of the relative minority among the classes, we propose to generate more samples in safer area or toward the relative majority with the larger size.

Suppose $|C_m|$ is below the average size, for a given sample $x_i \in C_m$, its oversampling weight is defined as:

$$W(x_i) = \frac{\sum_{C_n \in D, m \neq n} \left(\frac{|C_m|}{|C_n|} \cdot SI_{(C_n, C_m)}(x_i) \right)}{DI_{C_m}(x_i)} \quad (6)$$

To make the size of C_m reach the average, the roulette wheel operator repeatedly selects the synthesis seed through probabilities obtained by the oversampling weight. Thus, (1) safe samples locating in the dense area with the same class have higher priority to be selected as synthesis seeds than the samples far from the dense area. (2) The overlapping samples interweaving with the relative majority samples have priority to be selected as synthesis seeds than those interweaving with the relative minority samples, avoiding weakening the importance of the relative minority. (3) The overlapping samples

interweaving with the relative majority with a larger size have higher possibility than those interweaving with the relative majority with a smaller size. (4) The isolated samples' selected possibility is proportional to the low SI with its class and high DI with a different class, which depends on the data distribution.

To further strengthen the importance of relative minority and avoid the new unexpected samples negatively affecting the learning performance, we propose to generate new samples under the constraint of the relative majority and minority.

For a given synthesis seed $x_i \in C_m$, let x_j be the closet sample of the same class and y be the closet sample of different classes. An Euclidean ball of x_i representing the synthesis range of new samples is obtained as:

$$B(x_i, r) = \{x \mid \|x, x_i\|_2 \leq r\}, \quad (7)$$

where $\|\cdot\|_2$ denotes Euclidean norm and r depends on the distance of x_i to x_j and y , as Fig. 4 shown.

- (a) $r = d(x_i, y)$, if y is a relative majority and x_j is closer than y .
- (b) $r = 1/2 * d(x_i, y)$, if y is a relative majority and closer than x_j .
- (c) $r = d(x_i, x_j)$, if y is a relative minority and x_j is closer than y .
- (d) r is in-existence, if y is a relative minority and closer than x_j .

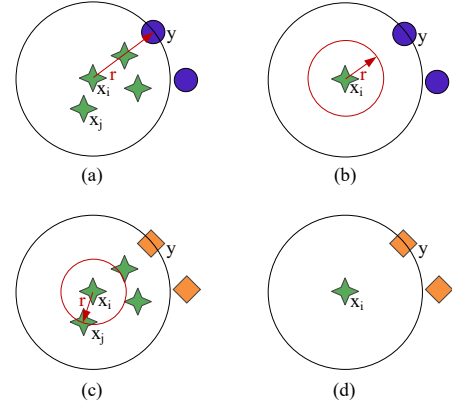


Fig. 4. Sketch map of new sample synthesis process, where the stars are the relative minority to the circles and the relative majority to the rhombuses.

The synthesis strategies in cases (a) and (b) aim to strengthen the importance of the relative minority, and strategies in cases (c) and (d) aim to avoid expending the negative affect of the relative majority.

D. Computational Complexity Analysis

To analyze the main computational complexity of SHSAMPLER, we define the total number of classes and samples as N and T , respectively. The sample numbers of each class are $\{t_1, t_2, \dots, t_N\}$. Suppose the number of features as d , and the complexity of calculating the distance of two samples is $O(d)$. In general, the computational complexity of SHSAMPLER can be analyzed by three components:

Algorithm 1: SHSampler

Input : Training dataset D ;
Output: New training dataset D' .

```
1 Initialize:  $S_{del}=\emptyset$ ;  $S_{syn}=\emptyset$ ; //  $S_{del}$ : Set of samples to
   be eliminated,  $S_{syn}$ : Synthetic sample set.
2  $average = (\sum_{i=1}^N |C_i|)/N$ ;
3 for  $C_m, C_n \in D$  do
4   Calculate  $m_{(C_m, C_n)}$  by (3);
5 for  $C_m \in D$  do
6   Calculate  $d_m$  by (1);
7   for  $x_i \in C_m$  do
8     Calculate  $DI_{C_m}(x_i)$  by (2);
9     for  $C_n \in D$  do
10       $\alpha_{(C_m, C_n)} = d_m + m_{(C_m, C_n)}$ ;
11      Calculate  $SI_{(C_m, C_n)}(x_i)$  by (4);
12 for  $C_m \in D$  do
13   Normalize( $SI_{C_m}(x \in C_m)$ );
14   for  $C_n \in D$  do
15     Normalize( $DI_{(C_m, C_n)}(x \in C_m)$ );
16 for  $C_m \in D$  do
17   if  $|C_i| > average$  then
18     for  $x_i \in C_m$  do
19       Calculate  $W(x_i)$  by (5);
20      $S_{del} \leftarrow S_{del} \cup (|C_i| - average)$  undersampled
       samples by the roulette wheel selection
       operator;
21  $D' \leftarrow D - S_{del}$ ;
22  $D_{minor} \leftarrow$  classes with lower number than average
   and sorted by the class size in ascending order;
23 for  $C_m \in D_{minor}$  do
24   for  $X_i \in C_m$  do
25     Calculate  $W(x_i)$  by (6);
26    $syn\_num \leftarrow |C_m| - average$ ;
27   while  $syn\_num > 0$  do
28     Select a synthesis seed  $x_i$  with the roulette
       wheel selection operator;
29     Synthesize a new sample  $x_j$  in
        $B(x_i, r) = \{x \mid \|x, x_i\|_2 \leq r\}$ ;
30      $S_{syn} \leftarrow S_{syn} \cup x_j$ ;
31      $syn\_num \leftarrow syn\_num - 1$ ;
32  $D' \leftarrow D' \cup S_{syn}$ .
```

(1) Complexity of similarity and dissimilarity estimation: At first, the compactness degree of each class and the separability degree between classes need to be calculated. For each class C_i , obtaining its centroid needs $O(t_i d)$, and calculating the samples of class C_i to the centroid also needs $O(t_i d)$. Thus the complexity of calculating the compactness degree of all classes is $O(t_1 d) + O(t_1 d) + \dots + (t_N d) + O(t_N d) = O(2Td)$. The

complexity of calculating the separability degree among every two classes is $O(((N-1)N/2)d)$. For the samples in datasets, the complexity of calculating the distance of the first sample to others is $O((T-1)d)$. Similarly, the second sample is $O((T-2)d)$. Therefore, the complexity of calculating the distance of all sample pairs is $O(((T-1)T/2)d)$. Finally, for each sample of class C_i with size t_i , estimating its dissimilarity with its class needs $O((t_i - 1))$. Similarly, estimating its similarity with other classes needs $O((T - t_i))$. Normalizing these values needs $O(t_i - 1 + T - t_i) = O(T - 1)$. Therefore, obtaining the similarity and dissimilarity values of all samples consumes $O((t_1 - 1) + (T - t_1) + T - 1 + \dots + (t_N - 1) + (T - t_N) + T - 1) = O(2(T - 1)N)$. Generally, the complexity of similarity and dissimilarity estimation needs $O(2Td) + O(((N-1)N/2)d) + O(((T-1)T/2)d) + O(2(T-1)N)$.

(2) Complexity of the relative majority weakening under-sampling: Suppose the number of classes above the average size is N_1 , and the total number of these classes is T_1 with the class size $\{t_1, t_2, \dots, t_{N_1}\}$. For each class C_i , the sampling weights of these samples are obtained with the complexity of $O(t_i)$. Undersampling these samples needs $O(t_i)$. Thus, the undersampling process needs $O(t_1) + \dots + O(t_{N_1}) = O(2T_1)$.

(3) Complexity of the relative minority strengthening over-sampling: Suppose the number of classes below the average size is N_2 , and the total number of these classes is T_2 with the class size $\{t_1, t_2, \dots, t_{N_2}\}$. Suppose the number of new samples to be generated is $K = \{k_1, k_2, \dots, k_{N_2}\}$. For each class C_i , the sampling weights of these samples are obtained with the complexity of $O(t_i)$. In one sampling iteration, selecting a synthetic seed needs $O(1)$, and determine the synthetic range r by detecting its nearest sample from the same class and different classes consumes $O(T - 1) + O(1)$. For convenience, synthesizing a new sample is supposed to have a complexity of $O(d)$. Thus, synthesizing a new sample needs $O(1) + O(T - 1) + O(1) + O(d) = O(T + 1 + d)$. Therefore, the oversampling process needs $O(k_1(T + 1 + d) + k_2(T + 1 + d) + \dots + k_{N_2}(T + 1 + d)) = O(K(T + 1 + d))$.

In summary, the total complexity is $O(2Td) + O(((N-1)N/2)d) + O(((T-1)T/2)d) + O(2(T-1)N) + O(2T_1) + O(K(T + 1 + d))$. As a result, the computational complexity of SHSampler can be simplified to $O(T^2 d)$. The pseudocode of SHSampler is presented in Algorithm 1.

IV. EXPERIMENTS

A. Experiment Settings

Datasets. To verify the performance, we have selected 20 multi-class imbalanced datasets as benchmarks from the KEEL repository [29]. The detailed information of these datasets can be seen in supplementary materials. To ensure thorough assessments, the datasets vary widely in the properties including imbalanced degree (Id) [30] and Fisher's discriminant ratio (Fdr) [31]. Fdr measures the overlapping degree of the dataset, and a lower Fdr means a higher overlapping degree.

Comparison methods. We consider 6 state-of-the-art approaches for multi-class imbalanced problem. These methods include (1) decomposition method: OVO-SMOTE (as OVO is

supposed to behave better than OVA [10], and SMOTE [32] is the most typical binary sampling method); (2) dedicated methods: MDO [15], SOUP [20], MC-RBO [18] and MC-CCR [19]; (3) a cost-sensitive-based method: Global-CS [33]. **Base classifiers.** We adopt three of the best-known classification algorithms as the base classifiers including: Decision Tree (DT), Random Forest (RF) and Naive Bayes (NB). These base classifiers are available in [34], and the default parameters remain unchanged. For training and testing of examined classifiers, we employ a stratified 5-fold cross validation [34]. **Evaluation metrics.** We evaluate the classification performance of multi-class imbalanced data with two commonly used metrics: extension of AUC (MAUC) [35] and multi-class G-measure (mGM) [2]. They are expressed as follows:

$$MAUC = \frac{2}{c(c-1)} \sum_{i < j} \frac{A(i, j) + A(j, i)}{2}, \quad (8)$$

$$mGM = \sqrt[c]{\prod_{i=1}^c recall_i}, \quad (9)$$

where c is the number of classes. $A(i, j)$ is the AUC [36] between class i and j calculated by the confusion matrix with the true class and predicted class. Meanwhile, $recall$ is the measure of completeness for each class [37].

B. Demonstration of Average Balancing Strategy

To avoid the potential overfitting or information loss caused by excessive oversampling or undersampling, we propose to obtain the average size of classes, undersample the classes above the size and oversample the classes below the size. To demonstrate the superiority of the average balancing strategy, we respectively conduct the other two strategies: (1) SHS_{min} , which only undersamples the classes to the minimum size; (2) SHS_{max} , which only oversamples the classes to the maximum size. Fig. 5 compares both strategies with SHSAMPLER on DT classifier.

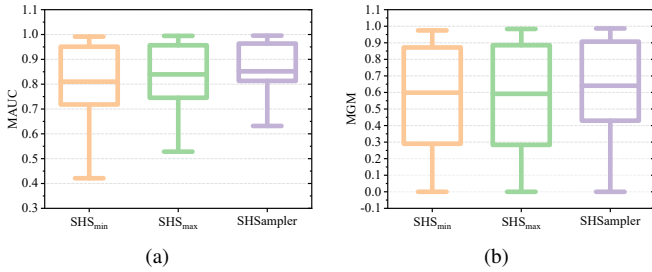


Fig. 5. Results of different balancing strategies on MAUC and mGM.

We can observe that: (1) SHSAMPLER outperforms SHS_{max} and SHS_{min} in terms of the mean values (represented by the center line). It demonstrates that the average balancing strategy is superior to the others. (2) SHS_{min} behaves worse than SHS_{max} . The reason may be that some datasets such as ‘ecoli’ and ‘lymphography’ have several extremely rare classes, and only undersampling classes to the minimum size possibly loses data information.

C. Ablation Analysis

The proposed SHSAMPLER is based on the hybrid framework with two sampling modules: a relative majority weakening undersampling and a relative minority strengthening oversampling. We denote the two components of SHSAMPLER as SHS_u and SHS_o , respectively. Fig. 6 compares the two components with SHSAMPLER on DT classifier. The ablation results reveal that: (1) both components behave worse than SHSAMPLER. The reason may be that, SHS_o can not avoid the negative effect of classes with large sizes, and SHS_u have limitation in highlighting the importance of classes with small sizes. (2) Both components have positive influence in improving the learning performance, and SHSAMPLER combining SHS_u and SHS_o achieves the best performance. The underlying reason may be that, SHS_u decreases the negative effect of the relative majorities by handling the difficulty factors, and SHS_o emphasises the relative minorities by supplementing safe samples, and both have interdependent effects to improve the classification performance.

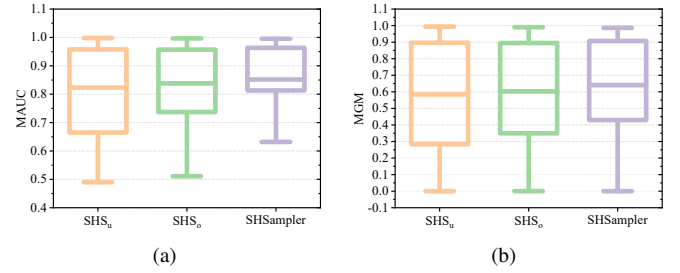


Fig. 6. Results of SHSAMPLER and its components on MAUC and mGM.

D. Comparison with Other Methods

To test the performance of SHSAMPLER, we compare it with 6 state-of-the-art methods on DT, RF and NB in this subsection. Tables I and II respectively show the results of MAUC and mGM over 20 datasets on DT classifier. The average and mean rank obtained by Friedman test [38] are also involved in the tables. The optimal results are recorded in bold. From Tables I and II, we can observe that SHSAMPLER outperforms the comparison methods in terms of MAUC and mGM, which verifies the superiority of SHSAMPLER. Specifically, SHSAMPLER gets the best average and mean rank, and the average of SHSAMPLER over 20 datasets gets 1.01% and 1.42% improvement on MAUC and mGM, respectively, compared to the second best method. Analogously, the results of RF and NB classifiers can be seen in supplementary materials.

We analyze the reasons in depth. OVO-SMOTE is a typical decomposition method and Global-CS is based on cost-sensitive. For the dedicated methods, MDO only generates new samples with the information of each class. MC-RBO and MC-CCR combine each class with its relative majorities, ignoring the relative minorities. Although SOUP considers the class interrelations, it is a neighboring-based method and conducts sampling in a simple removing and duplicating strategy. Compared with them, SHSAMPLER detects data difficulty factors based on a global sample estimation, and highlights

TABLE I
MAUC RESULTS ON DT CLASSIFIER

Dataset	OVO-SMO	Global-CS	MDO	MC-RBO	MC-CCR	SOUP	SHSampler
balance	0.6562	0.6166	0.6263	0.6180	0.6148	0.7863	0.6497
car	0.9832	0.9750	0.9794	0.9802	0.9779	0.9820	0.9755
cleveland	0.4186	0.5204	0.4766	0.4451	0.4942	0.5086	0.4961
ecoli	0.6387	0.5377	0.6394	0.5457	0.6092	0.5529	0.6317
glass	0.8839	0.8711	0.9006	0.9018	0.8822	0.9238	0.8615
hayes-roth	0.9202	0.9163	0.9202	0.9189	0.9420	0.9278	0.9353
led7digit	0.9549	0.9574	0.9566	0.9570	0.9340	0.9548	0.9546
lymphography	0.7003	0.7152	0.5568	0.7600	0.6780	0.6521	0.7650
newthyroid	0.9604	0.9562	0.9534	0.9587	0.9604	0.9555	0.9643
page-blocks	0.9496	0.9470	0.9533	0.9539	0.9570	0.9704	0.9710
PhishingData	0.9315	0.9444	0.9256	0.9295	0.9336	0.9391	0.9399
sensorReadings	0.9971	0.9972	0.9943	0.9970	0.9943	0.9923	0.9955
thyroid	0.9815	0.9623	0.9526	0.9521	0.9431	0.9868	0.9684
userknowledge	0.9381	0.8891	0.9063	0.9465	0.9206	0.8984	0.9422
vehicle	0.8819	0.8820	0.8841	0.8816	0.8867	0.8786	0.8898
vertebralColumn	0.8662	0.8532	0.8647	0.8819	0.8395	0.8479	0.8759
wine	0.9649	0.9643	0.9635	0.9592	0.9446	0.9551	0.9627
winequality	0.5135	0.5084	0.4906	0.4831	0.5540	0.5048	0.5118
yeast	0.7437	0.7061	0.7507	0.7573	0.7662	0.8039	0.8733
zoo	0.8369	0.8186	0.8436	0.8221	0.8162	0.8111	0.8693
average	0.8361	0.8269	0.8269	0.8325	0.8324	0.8416	0.8517
mean rank	3.48	4.50	4.43	4.15	4.50	4.15	2.80

TABLE II
MGM RESULTS ON DT CLASSIFIER

Dataset	OVO-SMO	Global-CS	MDO	MC-RBO	MC-CCR	SOUP	SHSampler
balance	0.3022	0.0418	0.0832	0.0832	0.0417	0.6304	0.2990
car	0.9468	0.9231	0.9363	0.9402	0.9321	0.9444	0.9263
cleveland	0.0000	0.1256	0.0000	0.0234	0.1016	0.0975	0.0591
ecoli	0.1182	0.1541	0.1400	0.1390	0.0000	0.1841	0.1506
glass	0.5393	0.4859	0.5813	0.5910	0.5582	0.6378	0.5125
hayes-roth	0.8308	0.8221	0.8308	0.8278	0.8738	0.8452	0.8594
led7digit	0.6800	0.6905	0.6866	0.6900	0.5763	0.6761	0.6746
lymphography	0.4370	0.5599	0.1943	0.5938	0.5610	0.4745	0.6432
newthyroid	0.9124	0.9058	0.9031	0.9110	0.9160	0.9081	0.9276
page-blocks	0.7955	0.7842	0.8234	0.8193	0.8324	0.8837	0.8883
PhishingData	0.8686	0.8911	0.8578	0.8649	0.8726	0.8830	0.8834
sensorReadings	0.9913	0.9915	0.9830	0.9910	0.9828	0.9770	0.9866
thyroid	0.9595	0.9121	0.8928	0.8911	0.8767	0.9718	0.9298
userknowledge	0.7525	0.5877	0.6563	0.7872	0.7187	0.6277	0.7713
vehicle	0.6724	0.6745	0.6833	0.6744	0.6826	0.6716	0.6964
vertebralColumn	0.7252	0.6851	0.7334	0.7502	0.6822	0.6793	0.7549
wine	0.9306	0.9304	0.9227	0.9194	0.8892	0.9100	0.9255
winequality	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
yeast	0.0505	0.0469	0.1080	0.0928	0.1601	0.1504	0.3464
zoo	0.4340	0.4797	0.4797	0.4797	0.3891	0.3758	0.5777
average	0.5973	0.5846	0.5748	0.6035	0.5824	0.6264	0.6406
mean rank	4.10	4.30	4.45	3.95	4.50	3.95	2.75

the relative minorities in mutual relationships with constraint undersampling and oversampling.

E. Statistical Test

To further compare the different algorithms, we use Bayesian analysis [39] to show the probability that one method is better than the other on DT, RF and NB classifiers. Specifically, Bayesian signed-rank test sets a value ('rope') of a region of practical equivalence, which indicates two methods tested on multiple datasets are considered equivalent if their performance difference is smaller than the 'rope'. Thus the test gives three probabilities: (1) one method is better than the other, (2) one method is worse than the other, or (3) they are in the 'rope'. Herein, we set the value of 'rope' as 0.5% with MAUC and mGM, and Fig. 7 shows the probability matrices.

We can observe that: (1) the probability of SHSampler exceeding comparison methods is mostly close or equal to 1 on DT classifier, which indicates that SHSampler achieves superior advantages on DT. (2) With respect to RF classifier, the performance of Global-CS, MC-RBO and MC-CCR on

MAUC is similar to SHSampler, outperforming the other methods. But the performance of these methods on mGM is worse than SHSampler. We can conclude that, although the general performance of Global-CS, MC-RBO and MC-CCR is competitive, the recognition completeness of each class by SHSampler is much better. (3) When it comes to NB classifier, SHSampler achieves superior advantages on MAUC. MC-RBO shows well performance on mGM, but its general performance on MAUC is much worse than SHSampler. To sum up, the proposed SHSampler achieves superior classification performance on three classifiers.

V. RELATED WORK

Recent researches for multi-class imbalanced problem can be mainly divided into decomposition methods [9], [10], [11], [12] and dedicated methods [15], [16], [17], [18], [19], [20].

The first group mostly decomposes a multi-class problem into sub-problems of binary-class. OVO (one-versus-one) [11] approaches train a classifier for each pair of classes with one selected as minority and the other as majority, while

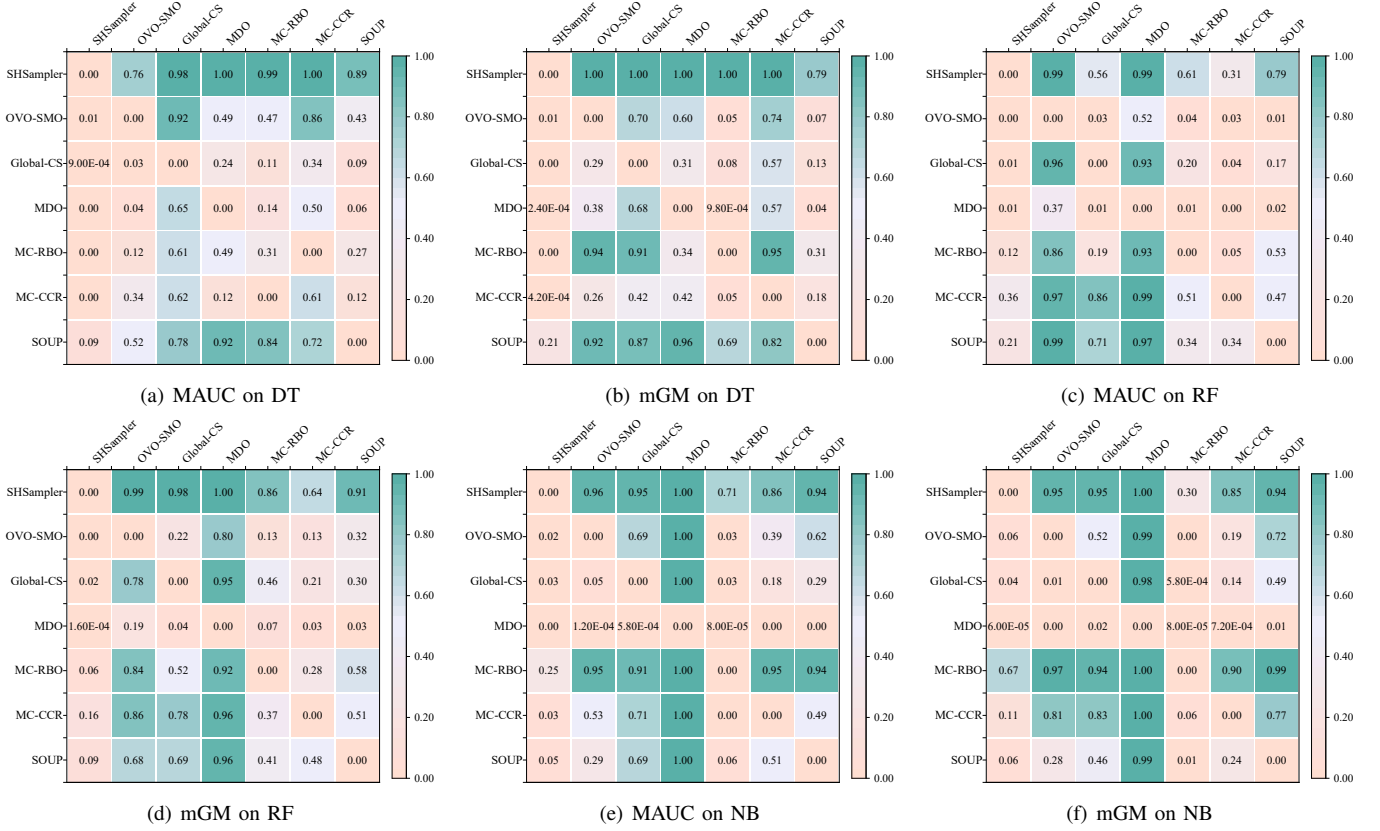


Fig. 7. Probability matrix obtained by the Bayesian signed-rank test on the three classifiers. The value in row_i and $column_j$ indicates the probability that $method_i$ exceeds $method_j$ by difference of 0.5%, and the color with darker cyan represents a higher value.

OVA (one-versus-all) [12] approaches build a classifier for each class considering it as minority, and the remaining samples constructs majority. Existing sampling methods such SMOTE [32] have been utilized in these approaches, and OVO methods generally have shown better behaviour [10].

The dedicated methods aim to modifying the original data distribution for better handling multi-class imbalanced data. For instance, MDO [15] generates new samples to persevere the covariance structure of the minority class by considering the Mahalanobis distance. SCUT [16] balances the number of multi-class by oversampling minorities with SMOTE [32] and undersampling majorities with cluster analysis. Considering the effect from different classes, SMOM [17] avoids over-generalization by selecting more safer neighbor directions for sample synthesizing. Other recent techniques such as MC-RBO [18] and MC-CCR [19] sample each class in terms of the compared majorities, MC-RBO uses potential function to generate samples and MC-CCR models a suitable over-sampling region with energy-based approach. Furthermore, SOUP [20] duplicates safe minority samples and removes difficult majority samples with exploiting information about the similarity of neighboring classes.

There are a few works handling multi-class imbalance problem based on cost-sensitive learning. For instance, Global-CS [33] examines the consistency of the class costs, then rebalances the classes with consistent costs or decomposes the multi-class with inconsistent costs.

VI. DISCUSSION

We discuss the superiorities and some limitations of our SHSAMPLER. Firstly, the high-level idea of SHSAMPLER is to emphasize the importance of the relative minorities by combating the mutual relationships with addressing data difficulty factors. SHSAMPLER has shown superiority to existing methods that have limitations in handling difficult mutuality. Secondly, SHSAMPLER utilizes the global information based sample estimation to distinguish data difficulty factors from safe samples, which is challenging and novel as well-known methods are mostly based on local information such as K-nearest neighbors. Besides, SHSAMPLER is parameter-free as it is adaptive to data distribution. However, multi-class imbalanced data is a complex problem, and in some cases, the classes may have extremely low sizes. In this scenario, similarly with most of the existing multi-class sampling techniques, our proposed seed selection strategy may also recurrently select these samples as seeds to generate new samples, thus our proposed method performs similarly with other compared methods.

VII. CONCLUSION

This work analyzes the multi-class imbalanced data associated with mutual relationships and data difficulty factors, and proposes a similarity-based hybrid sampling (SHSAMPLER) to address these problems. To highlight the importance of the relative minorities and mitigate the negative effect of data difficulty factors, SHSAMPLER provides a sample similarity and

dissimilarity estimation to identify data difficulty factors, and conducts an efficient hybrid sampling including a relative majority weakening undersampling and a relative minority strengthening oversampling. Experiments on 3 base classifiers and 20 datasets with various properties have shown the superiority of SHSampler. Since there are some extreme data distribution cases existing in multi-class imbalanced learning, our future work will focus on designing more robust strategies for the scenarios with extreme data distribution.

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TABLE III
SUPPLYMENTARY MATERIAL: BASIC INFORMATION OF MULTI-CLASS DATASETS

Dataset	#Instances	#Features	#Class	Class distributrion	Id	Fdr
balance	625	4	3	288/49/288	0.66	0.13
car	1728	6	4	65/69/384/1210	2.62	0.47
cleveland	297	13	5	160/54/35/35/13	3.44	0.06
ecoli	336	7	8	143/77/2/2/35/20/5/52	4.56	0.36
glass	214	9	6	70/76/17/13/9/29	3.42	0.08
hayes-roth	160	4	3	65/64/31	0.36	0.10
led7digit	500	7	10	45/37/51/57/52/47/57/53/49	3.08	2.21
lymphography	148	18	4	2/81/61/4	1.77	1.02
newthyroid	215	5	3	150/35/30	1.55	3.49
page-blocks	5473	10	5	4913/329/28/88/115	1.89	0.87
PhishingData	1353	9	3	103/548/702	0.05	0.03
sensorReadings	5456	24	4	328/826/2097/2205	1.11	3.85
thyroid	720	21	3	17/37/666	3.59	0.17
userknowledge	403	5	5	24/26/102/122/129	5.54	0.20
vehicle	846	18	4	199/212/217/218	4.40	3.00
vertebralColumn	310	6	4	60/100/150	3.87	0.77
wine	178	13	3	59/71/48	0.69	0.45
winequality	6497	11	6	30/216/2138/2836/1079/1935	1.48	0.22
yeast	1484	8	10	244/429/463/44/51/163/35/30/20/5	3.67	0.02
zoo	101	16	7	41/13/10/20/8/5/4	3.67	0.03

TABLE IV
SUPPLYMENTARY MATERIAL: RESULTS ON RF CLASSIFIER

Metric	AUC							mGM						
	OVO-SMO	Global-CS	MDO	MC-RBO	MC-CCR	SOUP	SHSampler	OVO-SMO	Global-CS	MDO	MC-RBO	MC-CCR	SOUP	SHSampler
balance	0.6834	0.6372	0.6288	0.6362	0.6420	0.7779	0.6323	0.3722	0.1637	0.0411	0.1262	0.1409	0.5968	0.1351
car	0.9471	0.9555	0.9454	0.9648	0.9637	0.9649	0.9542	0.8377	0.8611	0.8426	0.8957	0.8922	0.8934	0.8651
cleveland	0.4752	0.4982	0.4601	0.4646	0.5282	0.5270	0.5030	0.0295	0.0747	0.0000	0.0000	0.0911	0.0000	0.0928
ecoli	0.6493	0.7013	0.7418	0.6886	0.6327	0.6696	0.6746	0.0950	0.0968	0.1615	0.1026	0.0000	0.0000	0.1418
glass	0.7989	0.8634	0.8077	0.9433	0.8864	0.9096	0.8681	0.3360	0.4790	0.3291	0.7138	0.5710	0.6335	0.5147
hayes-roth	0.9198	0.9262	0.9287	0.9236	0.9220	0.8988	0.9200	0.8287	0.8376	0.8445	0.8332	0.8287	0.7812	0.8240
led7digit	0.9571	0.9544	0.9532	0.9557	0.9479	0.9557	0.9565	0.6908	0.6755	0.6697	0.6823	0.6247	0.6822	0.6814
lymphography	0.6671	0.7638	0.7483	0.7633	0.8521	0.7497	0.8383	0.4503	0.6187	0.5549	0.5217	0.7140	0.6321	0.6937
newthyroid	0.9550	0.9603	0.9712	0.9688	0.9682	0.9713	0.9661	0.9044	0.9068	0.9406	0.9331	0.9321	0.9400	0.9298
page-blocks	0.9687	0.9574	0.9620	0.9686	0.9677	0.9719	0.9793	0.8733	0.8246	0.8510	0.8748	0.8699	0.8910	0.9187
PhishingData	0.9210	0.9388	0.9312	0.9256	0.9306	0.9333	0.9426	0.8492	0.8818	0.8669	0.8584	0.8663	0.8720	0.8885
sensorReadings	0.9958	0.9961	0.9948	0.9961	0.9960	0.9896	0.9949	0.9873	0.9882	0.9843	0.9884	0.9881	0.9694	0.9847
thyroid	0.9656	0.9734	0.9361	0.9779	0.9820	0.9656	0.9868	0.9266	0.9423	0.8527	0.9535	0.9617	0.9181	0.9725
userknowledge	0.9328	0.9405	0.9225	0.9352	0.9441	0.9284	0.9338	0.7404	0.7601	0.7188	0.7352	0.7590	0.7174	0.7355
vehicle	0.8929	0.8903	0.9001	0.8947	0.8963	0.8879	0.8971	0.7020	0.6873	0.7100	0.6969	0.7046	0.6890	0.7015
vertebralColumn	0.8796	0.8897	0.8785	0.8988	0.8908	0.8871	0.8926	0.7620	0.7780	0.7585	0.7964	0.7755	0.7681	0.7846
wine	0.9873	0.9773	0.9764	0.9838	0.9852	0.9792	0.9877	0.9738	0.9551	0.9528	0.9678	0.9702	0.9579	0.9751
winequality	0.5123	0.5274	0.4953	0.5160	0.5386	0.5355	0.5522	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
yeast	0.6582	0.7535	0.7535	0.7647	0.7858	0.7647	0.7454	0.0000	0.0926	0.0000	0.0491	0.0497	0.1039	0.0461
zoo	0.8721	0.8186	0.7674	0.7729	0.7971	0.8424	0.8708	0.5891	0.3891	0.4323	0.3869	0.3891	0.4872	0.5960
average	0.8320	0.8462	0.8352	0.8472	0.8529	0.8555	0.8548	0.5974	0.6006	0.5756	0.6058	0.6064	0.6267	0.6241
mean rank	4.95	4.05	5.30	3.45	3.30	3.85	3.10	4.48	4.03	4.93	3.70	3.70	4.08	3.10

TABLE V
SUPPLYMENTARY MATERIAL: RESULTS ON NB CLASSIFIER

Metric	AUC							mGM						
	OVO-SMO	Global-CS	MDO	MC-RBO	MC-CCR	SOUP	SHSampler	OVO-SMO	Global-CS	MDO	MC-RBO	MC-CCR	SOUP	SHSampler
balance	0.8579	0.8616	0.6622	0.8694	0.8635	0.8289	0.8600	0.7222	0.7312	0.0426	0.7431	0.7324	0.6549	0.7311
car	0.9522	0.9522	0.9522	0.9522	0.9522	0.9522	0.9509	0.8579	0.8579	0.8579	0.8579	0.8579	0.8579	0.8540
cleveland	0.5084	0.4490	0.4311	0.5573	0.4613	0.5040	0.5859	0.0000	0.0000	0.0295	0.1236	0.0243	0.0594	0.1292
ecoli	0.3257	0.3032	0.2613	0.2442	0.2057	0.3271	0.3558	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
glass	0.6511	0.6306	0.6455	0.6894	0.6681	0.5745	0.6536	0.1270	0.1319	0.1667	0.1686	0.1803	0.0586	0.1575
hayes-roth	0.8235	0.8141	0.8244	0.8243	0.8189	0.7726	0.8564	0.6585	0.6398	0.6607	0.6664	0.6457	0.5957	0.7337
led7digit	0.8638	0.8742	0.9355	0.9017	0.9088	0.7793	0.9083	0.4715	0.4659	0.5983	0.5394	0.5410	0.1229	0.5470
lymphography	0.4733	0.4633	0.4581	0.5989	0.5780	0.5485	0.6470	0.1735	0.1526	0.1438	0.3874	0.3755	0.2660	0.5244
newthyroid	0.9731	0.9760	0.9644	0.9760	0.9654	0.9755	0.9642	0.9417	0.9478	0.9230	0.9478	0.9249	0.9467	0.9236
page-blocks	0.9182	0.9090	0.8847	0.8973	0.8869	0.9091	0.8906	0.7137	0.6859	0.5888	0.6608	0.6259	0.6890	0.6279
PhishingData	0.8087	0.8360	0.7502	0.8331	0.8250	0.7961	0.8182	0.6716	0.7096	0.5637	0.7045	0.6950	0.6514	0.6828
sensorReadings	0.8312	0.8247	0.6676	0.8264	0.8239	0.8411	0.8220	0.5995	0.5875	0.2755	0.5919	0.5866	0.6279	0.5815
thyroid	0.5941	0.5792	0.4824	0.6913	0.7301	0.5810	0.7647	0.3105	0.2917	0.1431	0.4402	0.4813	0.2945	0.5350
userknowledge	0.9183	0.9155	0.9094	0.9157	0.9178	0.9129	0.9136	0.6875	0.6798	0.6481	0.6806	0.6874	0.6892	0.6735
vehicle	0.7166	0.7189	0.7161	0.7177	0.7204	0.7468	0.7272	0.3802	0.3785	0.3806	0.3788	0.3848	0.4251	0.4011
vertebralColumn	0.8723	0.8690	0.8612	0.8796	0.8631	0.8791	0.8794	0.7591	0.7513	0.7427	0.7717	0.7411	0.7723	0.7716
wine	0.9934	0.9883	0.9810	0.9887	0.9883	0.9836	0.9873	0.9866	0.9763	0.9619	0.9771	0.9763	0.9669	0.9738
winequality	0.5791	0.6044	0.4981	0.6092	0.6157	0.6216	0.5958	0.1163	0.1166	0.0282	0.1211	0.1278	0.1316	0.0904
yeast	0.5932	0.5832	0.3445	0.7190	0.5989	0.6710	0.5603	0.0000	0.0000	0.0000	0.0246	0.0000	0.0000	0.0000
zoo	0.8373	0.8373	0.7054	0.7915	0.7387	0.8373	0.8529	0.3887	0.3887	0.0891	0.4319	0.2061	0.3868	0.4172
average	0.7546	0.7495	0.6968	0.7742	0.7565	0.7521	0.7797	0.4783	0.4746	0.3922	0.5096	0.4910	0.4598	0.5178
mean rank	3.74	4.26	6.07	2.76	3.76	4.02	3.38	4.00	4.43	5.57	2.79	3.62	4.00	3.60